

IdeaBlocks: Expressing and Reusing Exploratory Intents for Design Exploration with Generative AI

DaEun Choi
School of Computing
KAIST
Daejeon, Republic of Korea
daeun.choi@kaist.ac.kr

Kihoon Son
School of Computing
KAIST
Daejeon, Republic of Korea
kihoon.son@kaist.ac.kr

Jaesang Yu
School of Computing
KAIST
Daejeon, Republic of Korea
jaesangyu@kaist.ac.kr

HyunJoon Jung
Adobe
San Jose, California, USA
hjung@adobe.com

Juho Kim
School of Computing
KAIST
Daejeon, Republic of Korea
juhokim@kaist.ac.kr

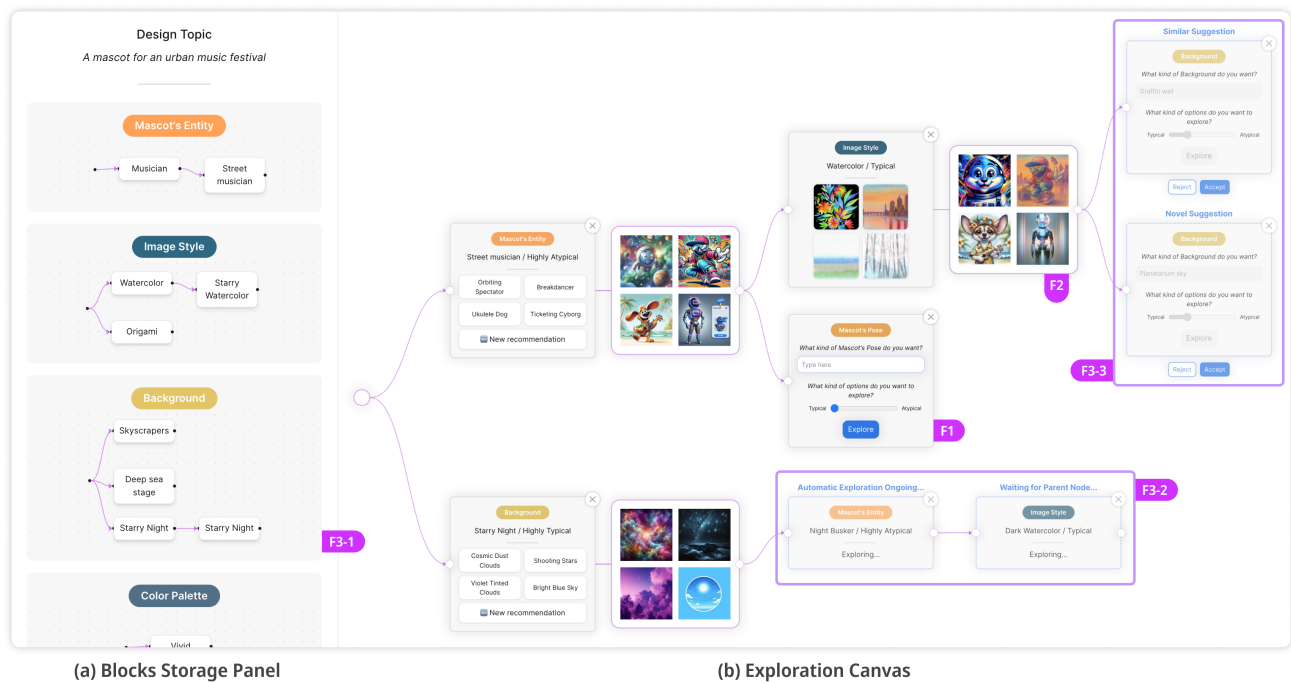


Figure 1: Overview of the IdeaBlocks system with (a) the *Blocks Storage Panel* and (b) the *Exploration Canvas*. Users begin by expressing their exploratory intent through structured inputs in *Exploration Blocks* (F1). These blocks can then be chained together to build on previous ideas across different properties (F2). Users can reuse individual blocks (F3-1), reuse entire exploration paths with optional adaptation (F3-2), or receive system-generated suggestions for the next direction (F3-3).

Abstract

Generative AI opens new possibilities for design exploration by rapidly generating images aligned with user goals. Yet, through

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3759001>

early user interviews, we observed several challenges that can limit broad and continuous exploration, such as difficulty articulating open-ended intent, fragmented workflows, and limited ways to build on previous ideas. To address these, we propose IdeaBlocks, where users can express their exploratory intents to generative AI with structured input and modularize them into *Exploration Blocks*. These blocks can be chained for continuous, non-linear exploration and reused across contexts, enabling broad exploration without losing creative momentum.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**.

Keywords

Creativity support tool, Design exploration, Generative AI, Graphic design

ACM Reference Format:

DaEun Choi, Kihoon Son, Jaesang Yu, HyunJoon Jung, and Juho Kim. 2025. IdeaBlocks: Expressing and Reusing Exploratory Intent for Design Exploration with Generative AI. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25), September 28–October 01, 2025, Busan, Republic of Korea*. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3746058.3759001>

1 Introduction

Design exploration is an essential step in creative process [2], where designers collect diverse references, discover what they like, and use those inspirations to develop optimal solutions [3–6]. Recent advancements in generative AI have introduced new opportunities to enhance design exploration. Image generation models, in particular, enable designers to rapidly produce tailored visual content from single prompts, supporting fast iteration and convergence when their design goal is clearly defined. These capabilities open new possibilities to accelerate the creative process and expand the design space.

Despite their potential, current interactions with these models often fall short in fostering creative divergence. To investigate this, we conducted an early user interview with 7 participants, focusing on their difficulties in expressing, evolving, and managing their exploration intentions when using generative models. Our findings revealed three key challenges: (1) participants struggled to express open-ended *exploratory intent*, especially in early stages when they had only rough ideas about areas of interest or direction, (2) they found it difficult to make incremental progress while preserving successful outcomes, leading to early fixation, and (3) there was limited support for reusing or evolving prior intentions, making it hard to revisit promising directions and maintain creative flow.

To address these challenges, we propose IdeaBlocks, a system that enables users to structure their exploratory intents into modular units called *Exploration Blocks* and interact with them on a canvas-based interface. Each *Exploration Block* encapsulates a user's area of interest, direction, and scope of exploration, providing a flexible way to interact with generative AI models. Users can freely manipulate and connect these blocks on a canvas to form non-linear exploration paths, supporting broad, continuous, and iterative idea development over time. IdeaBlocks also provides features for reusing previously created blocks and paths, enabling designers to efficiently revisit, adapt, and extend their ideas. By externalizing exploratory intents into modular and manipulable objects, IdeaBlocks supports users in organizing and navigating their creative thinking process in more breadth and efficiency.

2 IdeaBlocks

We implemented IdeaBlocks, a generative AI-based system supporting design exploration by enabling structured, iterative, and

non-linear workflows. At the core of IdeaBlocks is the concept of an *Exploration Block*—a modular unit that captures a user's *exploratory intent* for a specific design property, including both the direction and the desired level of typicality. This structure (Figure 2) includes the three key components of *exploratory intent* identified in our formative study: the area of interest (defined by selecting a property), the exploration direction (specified via a keyword input), and the exploration range (controlled through the typicality slider [1]). To address the challenge of articulating visual properties and abstract directions through text alone, IdeaBlocks augments user input with system-generated textual and visual suggestions, enabling users to refine their intent by selecting, modifying, or building on representative examples.

Through this structure, IdeaBlocks enables users to explicitly define their exploratory intent and modularize it into *Exploration Blocks*. These blocks can be chained to incrementally build the next exploration upon the previous blocks, supporting continuity throughout the process. Additionally, previously explored blocks and paths are stored as a reusable format, allowing users to efficiently revisit and evolve their ideas over multiple iterations.

2.1 Interface Walkthrough & Features

The system includes a canvas interface where users can make and interact with *Exploration Blocks* (Figure 1).

2.1.1 Generate a List of Properties. The exploration begins by entering a design topic. Based on this, IdeaBlocks suggests 8 design properties that are relevant to the topic, using GPT-4o. These properties are displayed in the left panel (Figure 1 (a)) and serve as the primary dimensions for exploration. Properties are classified as either text-based (e.g., "Character's Entity") or image-based (e.g., "Image Style").

2.1.2 F1. Expressing Exploratory Intent into Exploration Block. For each design property, users can create an *Exploration Block* to express their exploratory intent by entering a short keyword and adjusting a typicality slider (Figure 2). The keyword defines the direction of exploration (e.g., "Street musician"), and the typicality level (1 to 5) controls the range of the exploration. A lower typicality level (closer to 1) generates more aligned and conventional results, whereas a higher level (closer to 5) produces more divergent and unconventional outputs. This allows users to flexibly control the breadth of their exploration depending on their current needs.

Based on the user's inputs, the system generates four design suggestions, presented as either text or images, depending on the type of the selected property. Users can iteratively refine these suggestions by deleting or requesting similar or distant suggestions. The generated images corresponding to each suggestion are then displayed in *Result Block* next to their respective *Exploration Block*.

Each *Exploration Block*, along with its settings and outcomes, is stored and summarized in the left panel, allowing users to track their exploration process. These stored blocks can also be reused for further exploration, which will be described in Section 2.1.4.

2.1.3 F2. Chaining Blocks for Continuous Exploration. IdeaBlocks enables users to chain multiple *Exploration Blocks* on the canvas, allowing them to build upon previously explored ideas across different properties (Figure 3). When a new block is linked to an existing

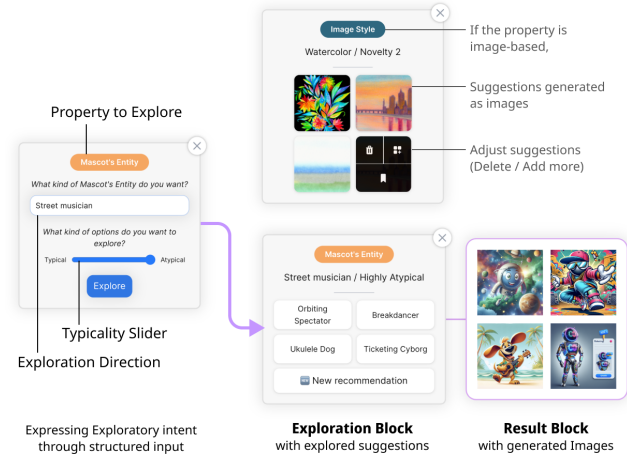


Figure 2: Expressing exploratory intent. Users create an *Exploration Block* for the specific design property they want to explore, enter the exploration direction via a keyword, and adjust the typicality level to control the range of the generated suggestions. IdeaBlocks generates four suggestions, either in text or image, depending on the property type. Users can refine these suggestions and review the resulting images in the adjacent *Result Block*.

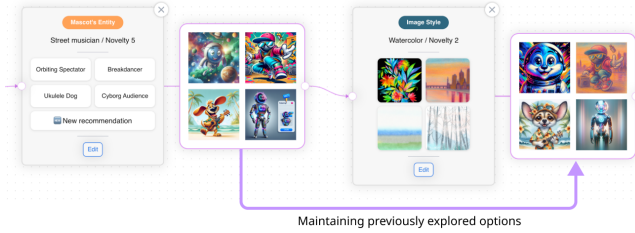


Figure 3: Chaining *Exploration Blocks* to maintain continuity of the exploration. Users link blocks to build on prior exploration, and each *Exploration Block* and *Result Block* becomes the context for generating the next results, allowing users to maintain previously explored intentions while exploring new properties.

one, the previously explored properties, suggestions, and their corresponding generated images serve as input context for the next stage of exploration. Therefore, the images generated in the next *Result Blocks* reflect not only the current intent but also the accumulated exploration history. This chaining mechanism preserves continuity in the creative direction and supports step-by-step expansion across multiple properties. As users chain multiple blocks into one block, those are shown as a branch, helping users to explore in a parallel way and easily keep track of how their ideas have evolved.

2.1.4 F3. Reusing Previous Intentions. To support efficient and flexible exploration, IdeaBlocks provides multiple features that enable users to revisit and reuse earlier exploratory intents.

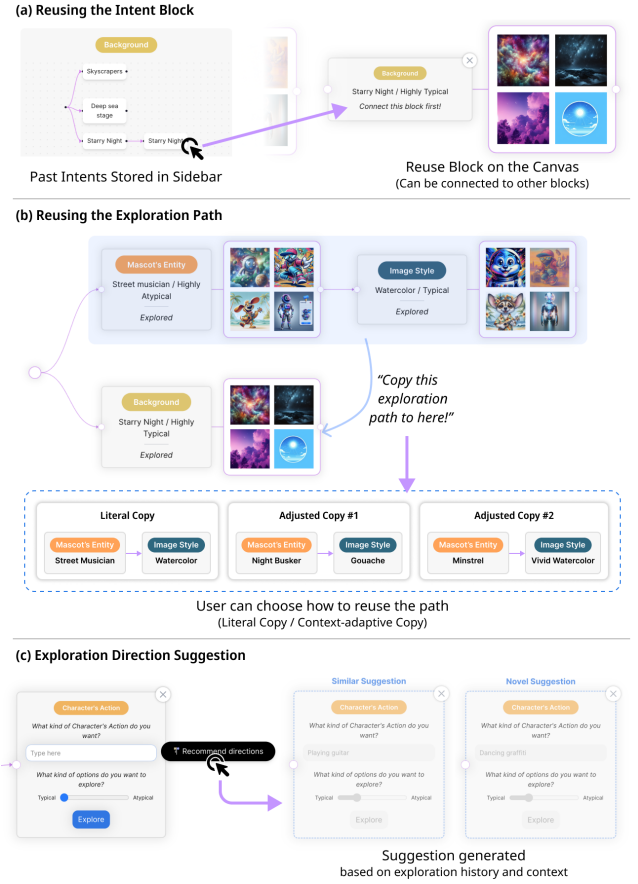


Figure 4: IdeaBlocks's three features for reusing prior exploration. (a) Users can reuse a previously created *Exploration Block* from the sidebar, placing it on the canvas with its original settings. (b) Users can copy and paste the entire *Exploration Paths*, choosing between literal copy and context-adaptive versions that are updated based on the new context. (c) Users can request system-generated direction suggestions that offer both familiar and novel paths based on exploration history and current context.

First, users can retrieve previously created *Exploration Blocks* from the left sidebar, where all past exploration histories are stored (Figure 4 (a)). While users explore different properties in the canvas, IdeaBlocks generates a graph in the left panel visualizing how their exploration direction evolves for each property. Selecting a block from this graph places a new copy onto the canvas with its original settings preserved. Users can then freely connect this copied block to new exploration paths, allowing them to extend or re-contextualize earlier ideas.

Second, users can reuse an entire sequence of previously created blocks by copying and pasting an exploration path (Figure 4 (b)). When users select a point in the canvas and choose a path to reuse, they are given two options: (1) a literal copy that replicates all blocks as-is, or (2) a context-adaptive copy that adjusts the contents of blocks based on their new context (i.e., the preceding blocks in the

current path). Once pasted, the system automatically creates blocks with appropriate settings and continues the exploration process with those blocks automatically. This enables users to efficiently branch off from a new starting point while preserving the structure of a familiar design trajectory.

Finally, users can draw on past explorations indirectly through IdeaBlocks's recommendations for exploration directions (Figure 4 (c)). When a new *Exploration Block* is created, users can receive two suggested inputs for the direction field: one that aligns closely with their previous explorations, and another that introduces a more novel idea. This feature offers a lightweight complement to the intentional reuse of past exploration by presenting context-aware suggestions based on prior directions and the current design context, helping users maintain continuity and spark new ideas.

Acknowledgments

This work was supported by Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No.2021-0-01347, Video Interaction Technologies Using Object-Oriented Video Modeling).

References

- [1] DaEun Choi, Kihoon Son, HyunJoon Jung, and Juho Kim. 2025. Expandora: Broadening Design Exploration with Text-to-Image Model. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 232, 10 pages. doi:10.1145/3706599.3720189
- [2] Claudia Eckert and Martin Stacey. 2000. Sources of inspiration: a language of design. *Design studies* 21, 5 (2000), 523–538.
- [3] Scarlett R. Herring, Chia-Chen Chang, Jesse Krantzler, and Brian P. Bailey. 2009. Getting Inspired! Understanding How and Why Examples Are Used in Creative Design Practice. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (*CHI '09*). Association for Computing Machinery, New York, NY, USA, 87–96. doi:10.1145/1518701.1518717
- [4] Brian Lee, Savil Srivastava, Ranjitha Kumar, Ronen Brafman, and Scott R. Klemmer. 2010. Designing with Interactive Example Galleries. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (*CHI '10*). Association for Computing Machinery, New York, NY, USA, 2257–2266. doi:10.1145/1753326.1753667
- [5] Felix Müller-Wienbergen, Oliver Müller, Stefan Seidel, and Jörg Becker. 2011. Leaving the beaten tracks in creative work—A design theory for systems that support convergent and divergent thinking. *Journal of the Association for Information Systems* 12, 11 (2011), 2.
- [6] Daniel Ritchie, Ankita Arvind Kejriwal, and Scott R. Klemmer. 2011. D.Tour: Style-Based Exploration of Design Example Galleries. In *Proceedings of the 24th Annual ACM Symposium on User Interface Software and Technology* (Santa Barbara, California, USA) (*UIST '11*). Association for Computing Machinery, New York, NY, USA, 165–174. doi:10.1145/2047196.2047216